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Effects of Lifestyle Modification on Patients with Resistant Hypertension: Results of the TRIUMPH Randomized Clinical Trial

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Abstract

Background: Although lifestyle modifications generally are effective in lowering blood pressure (BP) among patients with unmedicated hypertension and in those treated with 1–2 antihypertensive agents, the value of exercise and diet for lowering BP in patients with resistant hypertension (RH) is unknown.

Methods: One hundred forty patients with RH (mean age 63 years; 48% female; 59% Black; 31% with diabetes; 21% with chronic kidney disease) were randomly assigned to a 4-month program of lifestyle modification (C-LIFE) including dietary counseling, behavioral weight management, and exercise; or a single counseling session providing standardized education and physician advice (SEPA). The primary endpoint was clinic systolic BP (SBP); secondary endpoints included 24-hour ambulatory BP and select CVD biomarkers including baroreflex sensitivity (BRS) to quantify the influence of the baroreflex on heart rate, high frequency heart rate variability (HF-HRV) to assess vagally-mediated modulation of heart rate, flow-mediated dilation (FMD) to evaluate endothelial function, pulse wave velocity (PWV) to assess arterial stiffness, and left ventricular mass (LV mass) to characterize LV structure.

Results: Between-group comparisons revealed that the reduction in clinic SBP was greater in C-LIFE (–12.5 [95% CI: –14.9, –10.2] mm Hg) compared with SEPA (–7.1 [–95% CI: 10.4, –3.7] mm Hg) ($P = .005$); 24-hour ambulatory SBP also was reduced in C-LIFE (–7.0 [95% CI: –8.5, –4.0] mm Hg), with no change in SEPA (–0.3 [95% CI: –4.0, 3.4] mm Hg) ($P = .001$).

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Conflicts of Interest Disclosures: None

Supplemental Materials:

Expanded Methods

Supplemental Tables I-II

Compared with SEPA, C-LIFE resulted in greater improvements in resting BRS (2.3 msec/mm Hg [1.3, 3.3] vs. -1.1 msec/mm Hg [95% CI: -2.5, 0.3], $P < .001$), HF-HRV (0.4 ln ms² [95% CI: 0.2, 0.6] vs -0.2 ln ms² [95% CI: -0.5, 0.1], $P < .001$), and FMD (0.3 % [-0.3, 1.0] vs. -1.4 % [-2.5, -0.3], $P = .022$). There were no between-group differences in PWV ($p = .958$) or LV mass ($p = .596$).

Conclusions: Diet and exercise can lower BP in patients with RH. A 4-month structured program of diet and exercise as adjunctive therapy delivered in a cardiac rehabilitation setting results in significant reductions in clinic and ambulatory BP and improvement in selected CVD biomarkers.

Clinical Trial registration: [NCT02342808](https://clinicaltrials.gov/show/study/NCT02342808); [https://clinicaltrials.gov](https://clinicaltrials.gov/show/study/NCT02342808) › show › [NCT02342808](https://clinicaltrials.gov/show/study/NCT02342808)

Keywords

resistant hypertension; exercise; DASH diet; cardiac rehabilitation

INTRODUCTION

Resistant hypertension (RH) is defined as blood pressure (BP) that remains above goal (i.e., systolic blood pressure [SBP] ≥ 130 mm Hg and/or diastolic blood pressure [DBP] ≥ 80 mm Hg), despite adherence to a regimen of three or more optimally dosed antihypertensive medications of different classes, one of which is a diuretic.¹ Although estimates vary, RH is believed to affect approximately 5% of the general population and 20–30% of hypertensive adults, with even higher rates among adults with risk factors for cardiovascular disease (CVD) such as diabetes or chronic kidney disease.^{2, 3} RH is a particularly important designation because it is associated with a higher prevalence of end-organ damage and a 50% greater risk for adverse CVD events, including stroke, myocardial infarction, and death, compared with controlled BP.⁴ Therefore, identifying effective treatments for lowering BP in patients with RH is timely and important.

The failure to lower BP adequately with established antihypertensive medications in RH patients has prompted the evaluation of medications not typically used for first-line therapy of hypertension, such as spironolactone and amiloride,^{5, 6} and the development of device-based interventions, most notably renal denervation.^{7–9}

Surprisingly, lifestyle modification has not been rigorously evaluated in patients with RH. Several studies provided suggestive evidence that changes in diet and physical activity have the potential to lower BP substantially in these individuals.^{10, 11} However, these studies were small, treatment was as short as one week, the dietary modification was delivered in a clinical research unit rather than a ‘real world’ setting, and exercise and diet were studied separately. Lifestyle modifications, including both exercise training and dietary modification, particularly weight loss and the DASH eating plan, are of proven efficacy in lowering BP in unmedicated patients with hypertension and are often recommended as the first step for treating high BP¹². However, recent reviews of the management of RH have noted that the efficacy of these lifestyle modifications in RH patients has not been established.¹³ It cannot be assumed that lifestyle modification would be effective

in RH patients with a pathophysiology that renders them refractory to the most effective pharmacologic treatments. It is uncertain as to whether patients who are resistant to medications that are effective for the majority of patients with hypertension would benefit from lifestyle changes. The Treating Resistant hypertension Using lifestyle Modification to Promote Health (TRIUMPH) randomized clinical trial¹⁴ was designed to examine this issue by comparing a 4-month combined diet and exercise intervention, delivered in a cardiac rehabilitation setting, to a single educational session providing the same lifestyle prescription along with written guidelines to achieve specified exercise, weight loss, and nutritional goals.

METHODS

Trial Overview.

TRIUMPH was a randomized clinical trial (RCT) designed to evaluate whether an intensive, medically supervised lifestyle intervention, compared with an education and advice condition, can achieve clinically significant BP reductions in patients with RH. Enrollment began in June 2015 and ended in October 2019. The trial was approved by the institutional review board at Duke University School of Medicine and written informed consent was obtained from all participants. An independent data and safety monitoring board oversaw the conduct of the study and reviewed study progress and safety data. The study is registered at www.clinicaltrials.gov (NCT02342808) and the trial methodology has been previously reported.¹⁴ All data and a detailed description of the assessment procedures and data analyses will be made available upon reasonable request for the purpose of replicating these results. One hundred forty patients with RH were randomized with 2:1 allocation to either a Center-based Lifestyle Intervention (C-LIFE) or Standardized Education and Physician Advice (SEPA). The primary endpoint was clinic SBP; secondary endpoints included clinic DBP, 24-hour ambulatory BP, body weight, aerobic fitness, and selected markers of CVD risk including baroreflex sensitivity (BRS), high frequency heart rate variability (HF-HRV), pulse wave velocity (PWV), flow-mediated dilation (FMD) of the brachial artery, and left ventricular mass (LVM) and relative wall thickness. All end-points were measured at baseline and again after the completion of the 4-month interventions.

Participants

Patients with RH, defined as treatment for at least 6 weeks with 3 or more antihypertensive medications of different classes, including a diuretic, with clinic SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg (modified from 140/90 mm Hg in November 2017), or the need for 4 or more drugs to achieve SBP ≤ 130 mm Hg and DBP ≤ 80 mm Hg, with SBP ≥ 120 mm Hg were eligible. Additional inclusion criteria included body mass index (BMI) ≥ 25 kg/m², lack of regular moderate or vigorous physical activity, and age 35–80 years. Exclusion criteria included known secondary hypertension, estimated glomerular filtration rate (eGFR) < 40 ml/min/1.73 m², moderate-severe ischemic heart disease, diabetes requiring insulin, and major psychiatric disorder or substance dependence, including alcohol consumption > 14 drinks/week.

Screening Procedures and Baseline Laboratory Studies

All eligible patients had a series of BP measurements performed according to JNC 7 guidelines¹⁵, as described below. Serum creatinine and eGFR (calculated by the CKD-EPI equation), electrolytes, TSH, plasma renin activity, and serum aldosterone were measured on study entry.

Medication Adherence

Treatment regimens were documented by having participants bring their BP medications to the research clinic and were verified with referring physicians. Adherence was confirmed by self-report and objectively using the Medication Event Monitoring System (or MEMS®) bottle cap for 10 days.¹⁶ The intensity of pharmacotherapy was quantified as the number of antihypertensive medications, and as the Daily Defined Dose (DDD), a measure of drug amount that takes into account both the number of medications and their doses.¹⁷

Treatment Conditions

Participants were randomized to one of two 4-month treatment arms and were encouraged to maintain their prescribed antihypertensive medications at the discretion of their physicians. The prescribed lifestyle changes were essentially the same for participants in both 4-month treatment arms, with the difference being the intensity of the delivery of the interventions.

(1) Center-based Lifestyle Intervention (C-LIFE): Participants received instruction from a nutritionist on the DASH diet with caloric and sodium restriction (≤ 2300 mg/day) and underwent weekly 45-minute group counseling sessions delivered by a clinical psychologist that emphasized changes in the initiation of eating behavior, individualized problem-solving, and maintenance of long-term behavior change. Participants also exercised at a cardiac rehabilitation facility 3 times per week for 30–45 min at a level of 70–85% of their initial heart rate reserve (See Expanded Methods in Supplementary Materials for more details about the C-LIFE intervention).

(2) Medical Management with Standardized Education and Physician Advice (SEPA): Patients received a 1-hr educational session on BP management delivered by a health educator along with a workbook in which they received an individualized diet and exercise program including instruction in the DASH diet with caloric restriction and the same exercise prescription received by C-LIFE.

Outcome measures

Blood pressure.—The primary outcome measure was clinic SBP. Clinic-measured BP was determined according to JNC-7 guidelines. After 5 minutes of quiet rest, 4 seated BP readings, each 2 minutes apart, were obtained using an Accutorr Plus BP monitor (Datascop, Mahwah, New Jersey) with the first reading discarded to provide an average of 3 clinic BP measures per session. This clinic BP measurement protocol was repeated on 3 sessions over a 3 to 4-week period and the same protocol was repeated following the 4-month interventions. In this manner, the clinic BP values represented the average of nine BP readings obtained over a 3-week period. Clinic DBP was a secondary outcome measure.

Ambulatory SBP and DBP averaged over 24 hours were secondary BP endpoints. To assess BP during a typical day, participants wore an Oscar 2 (Suntech Medical Inc, Raleigh, North Carolina) ambulatory BP monitor (ABPM).¹⁸ The Oscar 2 was programmed to record BP measurements 3 times per hour throughout the waking hours and 2 times per hour during sleep. At least 70% of planned readings were required for an ambulatory BP record to be considered acceptable.¹⁹

Nutritional and Weight Assessment.—An independent assessment of dietary and nutritional content was obtained by 2 separate self-report measures of diet: a retrospective food frequency questionnaire (FFQ) requiring participants to recall typical consumption during a 4-week period²⁰ and a 2-day (weekend/weekday) food diary.²¹ The FFQ was analyzed by NutritionQuest (Berkeley, California), and the diary data were analyzed using the Automated Self-Administered 24-hour (ASA 24[®]) Dietary Assessment Tool (<https://epi.grants.cancer.gov/asa24/>). An overall DASH eating plan score was quantified using a scoring algorithm used previously.²² In addition, sodium and potassium intake were estimated from urinary excretion over a 24-hour period.²³ Body weight was determined by a calibrated digital scale (Detecto; Cardinal Scale Manufacturing Co, Webb City, Missouri).

Cardiorespiratory Fitness.—Participants underwent a maximal graded exercise treadmill test in which workloads were increased at a rate of 1 metabolic equivalent per minute.²⁴ Expired air was collected by mouthpiece for quantification of minute ventilation, oxygen consumption, and carbon dioxide production with the Parvo Medics TrueOne measurement system (model 2400; Parvo Medics, Sandy, Utah).

CVD and Metabolic Biomarkers.—Secondary outcomes included measurement of BRS to quantify the influence of the baroreflex on heart rate,²⁵ PWV to assess arterial stiffness,²⁶ HF-HRV to assess vagally mediated modulation of heart rate,²⁷ FMD to evaluate endothelial function^{28, 29}, LVM and relative wall thickness to characterize left ventricular structure,³⁰ C-reactive protein, and blood markers reflecting metabolic function (e.g., insulin, glucose, and lipids). (See Expanded Methods for more details of the CVD biomarker assessment procedures).

Randomization

Participants were randomized to C-LIFE and SEPA in clusters of 3 to 5 participants when possible. The size of the group was determined by how many eligible participants were available to be randomized within 4–5 weeks of the completion of their baseline BP assessments. Participants were provided treatment group assignments in sealed envelopes; staff members performing assessments were blinded to participants' group assignments.

Statistical Analysis

All analyses were conducted using SAS 9.4. To examine changes in the primary outcome, clinic SBP, general linear models were used with post-treatment SBP as the outcome (4-months), treatment group assignment (C-LIFE vs SEPA) as a between-subjects factor, and race, sex, age, diagnosis of diabetes or chronic kidney disease (CKD), baseline medication adherence, and pretreatment clinic SBP as covariates, selected *a priori*. An ancillary analysis

was conducted to evaluate the possible role of change in BP medication on the treatment effect by adding the DDD measure as a covariate in the models testing treatment group effects. Clinic DBP, ambulatory 24-hr BP, and daytime and nighttime BPs were analyzed using a parallel approach as the primary analyses, as were analyses of lifestyle changes following treatment including body weight, aerobic capacity, and dietary intake. Changes from pre-to-post treatment are expressed as model-adjusted least squared means with the accompanying 95% confidence interval (95% CI). For secondary analyses of multiple CVD and metabolic biomarkers, a mean rank composite was used to control for type I error.³¹ Use of such ‘gatekeeper’ approaches also lessens the influence of any outliers by restricting the distribution of outcomes using rank values (1 to 140) separately at baseline and post-treatment. All analyses of primary and secondary outcomes were performed following the intent-to-treat principle using Markov Chain multiple imputation analyses with 100 imputations (PROC MI and PROC MIANALYZE in SAS 9.4). Missing data were infrequent, with the vast majority of assessments having $\geq 93\%$ complete data prior to imputation and 91% of participants providing complete ambulatory BP data. Missing data were most common for vascular ultrasound assessments of FMD (16%) and BRS (26%) because of the COVID-19 pandemic during which in-person laboratory visits were not permitted. Unless otherwise indicated, all analyses represent between-group comparisons.

RESULTS

The CONSORT diagram for study participants is shown in Figure 1. As illustrated in the diagram, 506 patients were initially contacted, 266 screened, and 140 met the eligibility criteria and were randomized to one of the interventions. Of the 140 randomized participants, 67 were enrolled after the 2017 ACC/AHA BP guidelines were published and the threshold BP for inclusion was lowered to clinic SBP ≥ 130 mm Hg or DBP ≥ 80 mm Hg; 21 of these would not have qualified using the prior BP criteria. Ninety individuals were randomized to C-LIFE and 50 to SEPA. During the trial, only four patients dropped out (all from SEPA). However, due to the COVID-19 pandemic, three SEPA participants were unable to complete portions of their post-treatment assessments involving in-person laboratory biomarker measures. For these patients, clinic BPs were obtained remotely from home measurements and ambulatory BP devices were mailed to participants for self-application and monitoring. Using intention-to-treat analyses, all 140 participants were retained in the final analyses.

Sample Characteristics

Background and clinical characteristics of the sample are presented in Table 1. Men and women were equally represented and the sex distribution was similar in the two groups. Participants averaged 63 years of age and were more likely to be African American (59%) and obese (mean BMI = 36.0 [SD = 5.7] kg/m²). Almost a third had diabetes, and 24% had a history of CKD. No participants had evidence of clinically significant hyper- or hypothyroidism or unexplained hypokalemia. Fourteen patients had suppressed plasma renin activity and serum aldosterone > 10 ng/dl. On average, participants were prescribed 3.5 (SD = 0.7) antihypertensive medications.

Adherence to Treatment Protocol

Examination of C-LIFE participants' attendance to exercise training and dietary classes revealed excellent adherence. Participants attended a median of 94% of their DASH diet classes (IRQ: 81%, 94%) (Mean = 88.4 [SD = 11.3]) and 89% of structured exercise sessions (IRQ: 84%, 96%) (Mean = 88.3 [SD = 11.6]), and were in their target HR training range 94% of the time (IRQ: 82%, 99%) (Mean = 86.4 [SD = 16.7]).

Adherence to Medication Regimen

Participants in both groups exhibited excellent adherence to their antihypertensive medication regimens. Examination of MEMS data demonstrated a high degree of adherence at baseline (mean percentage medications taken: C-LIFE = 97% [SD = 7], SEPA = 97% [SD = 6]; mean percent medications taken within appropriate window: C-LIFE = 94% [SD = 11], SEPA = 95% [SD = 8]). Following the completion of the intervention, MEMS documented that participants in both conditions maintained their high levels of adherence for medications taken (mean percent medications taken: C-LIFE = 95% [SD = 9], SEPA = 96% [SD = 7]; $P = .309$) and for medications taken within the prescribed time window, with no treatment group differences (mean percent medications taken within appropriate window: C-LIFE = 93% [SD = 12], SEPA = 93% [SD = 12]; $P = .489$).

Lifestyle Outcomes

Examination of changes in aerobic fitness and functional capacity revealed greater improvements in peak VO_2 in the C-LIFE condition (14.8% VO_2 improvement [95% CI: 11.0, 18.6]) compared with SEPA (3.4% VO_2 improvement [95% CI: -2.3, 9.2]) ($P = .002$), as well as greater exercise duration (Table 2). Increased physical activity also was observed by actigraphy-determined total steps, with greater activity documented in C-LIFE (971 steps per day increase [95% CI: 568, 1374]) compared with minimal changes in SEPA (14 steps per day increase [95% CI: -580, 608]; $P < .01$).

Examination of weight changes and dietary composition revealed a similar pattern. C-LIFE participants exhibited substantial weight loss (-15.3 lbs. [95% CI: -17.2, -13.3]) compared with significant, albeit more modest, weight loss in the SEPA condition (-8.5 lbs. [95% CI: -11.4, -5.6]) ($P < .001$). The C-LIFE group demonstrated greater reductions in urinary sodium excretion compared with SEPA (-308 mg/day [95%CI: -567, -50] vs. -187 mg/day [95% CI: -288, 662], $P = .007$), whereas urinary potassium was not differentially improved between groups (C-LIFE: 416 mg/day [95% CI: 286, 546] vs. SEPA: 292 mg/day [95% CI: 96, 488], $P = .382$). There were modest improvements in adherence to the DASH eating plan in both C-LIFE and SEPA (0.6 DASH score increase [95% CI: 0.3, 0.8] vs. 0.3 DASH score increase [95% CI: -0.1, 0.6], respectively, $P = .262$). Unadjusted pre-post values of lifestyle measures are provided in Table I in the Supplement.

There was no significant change in BP medication burden, as measured by the DDD, from baseline to post-treatment ($P = .627$), nor any evidence to suggest differential changes between treatment groups ($P = .842$ for time X treatment group interaction). At baseline, participants in C-LIFE were taking an average of 5.1 (SD = 2.3) standard DDDs of BP medications compared with 4.9 (SD = 1.7) in SEPA. Following treatment, the C-LIFE group

demonstrated modest reductions in BP medication burden (-0.15 DDDs [95% CI: -0.30 , 0.01]) compared with no changes in the SEPA group (0.00 DDDs [95% CI: -0.22 , 0.23]; $P = .277$). There were no post-intervention group differences in the number of antihypertensive medications (C-LIFE = 3.5 [0.8]; SEPA = 3.5 [0.8]; $P = .929$).

Clinic BP Outcomes

C-LIFE participants exhibited lower clinic SBP (126.8 [95% CI: 124.5 , 129.1] mm Hg), the primary outcome, compared with SEPA (132.8 [95% CI: 129.5 , 136.1] mm Hg, $P = .005$). A similar improvement was noted for clinic DBP, with C-LIFE exhibiting lower DBP (73.2 [95% CI: 71.9 , 74.4] mm Hg) compared with SEPA (75.6 [95% CI: 73.8 , 77.4] mm Hg, $P = .034$). (Figure 2) Examination of clinic BP changes revealed greater reductions in clinic SBP in C-LIFE (-12.5 [95% CI: -14.9 , -10.2] mm Hg), compared with more modest reductions in SEPA (-7.1 [95% CI: -10.4 , -3.7] mm Hg) ($P = .005$), for a net difference in SBP reductions of -5.4 (95% CI: -9.7 , -1.2) mm Hg. (Figure 3) Changes in clinic SBP for individual participants in the two groups are illustrated in Figure 4. There was no evidence for treatment group interactions with age ($P = .751$), sex ($P = .455$), race ($P = .281$), or baseline SBP ($P = .440$). Similarly, treatment group effects were not differentially affected by the presence of diabetes ($P = .698$ for clinic SBP interaction) or CKD ($P = .745$ for clinic SBP interaction). Similar reductions were noted for clinic DBP, with moderate reductions in C-LIFE (-5.9 [95% CI: -7.2 , -4.7] mm Hg) compared with SEPA (-3.7 [95% CI: -5.6 , -1.8] mm Hg) ($P = .034$), for a net difference in DBP reductions of -2.2 (95% CI: -4.4 , 0.0) mm Hg. (Figures 2 and 3). BP $< 130/80$ mmHg was achieved in 59% of C-LIFE participants, compared with 38% of SEPA participants.

Ambulatory BP Outcomes

Post-intervention 24-hour ambulatory BPs were significantly lower for C-LIFE (126.4 [95% CI: 124.0 , 128.9] mm Hg/ 67.2 [95% CI: 65.8 , 68.6] mm Hg) compared with SEPA (133.2 [95% CI: 129.4 , 136.7]/ 70.8 [68.7, 72.9] mm Hg) ($P = .001$ for SBP and $P = .002$ for DBP) (Figure 2). Daytime SBP was lower in C-LIFE compared with SEPA ($P = .003$) (129.3 [95% CI: 126.8 , 131.8] v 136.5 [132.8, 140.2] mm Hg) as was nighttime SBP ($P = .047$) (118.5 [95% CI: 115.8 , 121.2] v 123.5 [95% CI: 119.2 , 127.6] mm Hg). Daytime DBP also was lower for C-LIFE compared with SEPA ($P = .013$) (69.7 [95% CI: 68.2 , 71.2] v 73.2 [95% CI: 71.1 , 75.5] mm Hg), while nighttime DBP was marginally lower for C-LIFE compared with SEPA ($P = .068$) (60.8 [95% CI: 59.4 , 62.3] v 63.0 [95% CI: 60.7 , 65.4] mm Hg).

Examination of changes in 24-hour ambulatory SBP revealed a significant reduction in C-LIFE (-7.0 [95% CI: -8.5 , -4.0] mm Hg), with no change in SEPA (-0.3 [95% CI: -4.0 , 3.4] mm Hg) ($P = .001$). (Figure 3). Similarly, reductions in 24-hour ambulatory DBP were greater in C-LIFE (-3.9 [95% CI: -5.3 , -2.5] mm Hg) compared with SEPA (0.3 [95% CI: -2.4 , 2.8] mm Hg) ($P = .002$). Changes in 24-hour ambulatory SBP for individual participants in the two groups are illustrated in Figure 4.

CVD Biomarker Outcomes

Examination of post-intervention changes in CVD risk biomarker (mean rank) revealed more favorable CVD biomarker profiles among C-LIFE participants compared with SEPA

($P = .027$). Explanatory follow-up analyses revealed that these improvements were driven primarily by favorable changes in BRS, HF-HRV, and FMD. Individuals in C-LIFE demonstrated greater improvements in resting BRS (2.3 msec/mmHg [95% CI: 1.3, 3.3]) compared with SEPA (-1.1 msec/mmHg [95% CI: -2.5 , 0.3]) ($P = .003$). Similarly, C-LIFE participants showed increased resting HF-HRV ($0.4 \ln \text{ms}^2$ [95% CI: 0.2, 0.6]) that was not observed in SEPA ($-0.2 \ln \text{ms}^2$ [95% CI: -0.5 , 0.1]) ($P = .025$). Finally, C-LIFE participants exhibited improvements in FMD (0.3 % [-0.3 , 1.0]) relative to SEPA participants (-1.4% [-2.5 , -0.3], $P = .022$). Group differences in the other individual metabolic biomarkers were not significant (Table 3). Unadjusted pre-post CVD biomarker values are provided in Table II in the Supplement.

DISCUSSION

This randomized clinical trial showed a benefit of an intensive lifestyle intervention for reducing both clinic and ambulatory BP in patients with RH. Although diet and exercise are generally well-established for the treatment of hypertension,^{32, 33} documentation of the antihypertensive benefits of lifestyle modifications in RH patients is almost nonexistent. Hypertension Canada's 2020 comprehensive review of the evidence regarding the treatment and management of RH noted that there were no RCTs that reported clinically relevant outcomes following health behavior modifications in RH.³⁴ A recent review by Carey¹³ noted that no studies had evaluated the effect of weight loss or the DASH diet in RH patients; two small studies, with 12 and 20 patients, respectively, showed BP lowering with sodium restriction,^{10, 35} and only two small studies demonstrated BP lowering with walking¹¹ or water-based exercises.³⁶ Until TRIUMPH, no studies had evaluated the effects of comprehensive lifestyle modification in RH. The combination of exercise, weight loss, and the DASH diet with sodium restriction is not only effective for unmedicated hypertensive patients, as had been demonstrated previously,³⁷ but also produces clinically significant BP lowering benefits in patients with RH. Moreover, the present findings provide strong support for the value of providing an intensive, structured intervention to achieve these benefits. While RH patients who received a 1-hour counseling session that included educational materials and advice for optimal BP management achieved significant clinic SBP reductions, no change was observed in ambulatory BP. The more intensive intervention delivered in a cardiac rehabilitation setting provided even greater lowering of clinic BP as well as a large reduction in ambulatory BP and an improvement in CVD biomarkers.

The BP reductions in TRIUMPH are comparable to those observed with antihypertensive medications³ and were achieved without contributing to pill burden and the risk of medication interactions and untoward side effects.³⁷ Participants in the C-LIFE intervention achieved more than a 12 mm reduction in clinic SBP, which was more than 5 mm Hg greater than the comparison group of patients who received the same lifestyle guidelines but had to achieve the behavioral changes on their own. Importantly, the majority of C-LIFE participants (59%) achieved the goal of clinic BP $< 130/80$ mm Hg after 4-months. Reductions in ambulatory BP also were greater for C-LIFE participants compared with SEPA controls, with C-LIFE participants showing a nearly 7 mm Hg greater reduction in 24-hour SBP, reflecting lowering of both daytime and nighttime pressures.

BP reductions of the magnitude observed in TRIUMPH are associated with improved cardiovascular outcomes in clinical trials with antihypertensive medications.³⁸, and are as large or larger than those observed in previous lifestyle intervention trials conducted in patients on one or two antihypertensive medications. In the TONE trial, for example, patients in the combined lifestyle condition achieved a clinic BP reduction of $-5.3/-3.4$ mm Hg,³⁹ and the 4-month lifestyle intervention employed in the ADAPT trial resulted in a reduction of $-4.1/-2.1$ mm Hg in ambulatory BP.⁴⁰ In the PATHWAY-2 study of patients with RH, patients achieved an average reduction in home SBP of -8.7 mm Hg with spironolactone, which was superior to placebo, bisoprolol, or doxazosin.⁵

It is difficult to compare the present findings with results from studies of renal denervation. The SYMPPLICITY HTN-2 trial generated considerable optimism by suggesting that percutaneous renal sympathetic denervation could lower clinic SBP by as much as 30 mm Hg in RH patients.⁷ However, a subsequent, more rigorously designed study of renal denervation proved inconclusive.⁸ In the recently published RADIANCE-HTN TRIO study⁹, among RH patients with ABPM data at baseline and after treatment, ultrasound-based renal denervation was associated with a reduction of -9.4 mm Hg in 24-hour ambulatory SBP compared with -2.9 mm Hg in the sham control group. These results are comparable to those observed in TRIUMPH (-7.0 [95% CI: $-8.5, -4.0$] mm Hg in C-LIFE and -0.3 [95% CI: $-4.0, 3.4$] mm Hg in SEPA).

In addition to evaluating the effects of lifestyle modifications on clinic and ambulatory BP, TRIUMPH examined changes in a constellation of cardiovascular biomarkers. C-LIFE participants achieved significant improvements in FMD, BRS and HF-HRV, all of which are associated with CVD risk⁴¹⁻⁴⁷ Although TRIUMPH was not powered to assess adverse CVD events, improvement in these surrogate endpoints suggests that lifestyle modifications may result in improved clinical outcomes^{41, 48}.

Limitations.

Because TRIUMPH was conducted at a single site, there may be concerns about the generalizability of these findings. However, the intervention was delivered at multiple cardiac rehabilitation facilities in central North Carolina, with broad representation of minorities and individuals with varied educational and cultural backgrounds. Thus, the beneficial effects of structured lifestyle interventions on BP and CVD risk factors are likely to be achieved across a broad range of patients. Only five TRIUMPH participants had SBP > 160 mm Hg, so the efficacy of lifestyle modifications in patients with severe RH or refractory hypertension could not be definitively established. However, the effects of the intervention were not moderated by baseline SBP, suggesting that lifestyle changes may be effective in patients treated with multiple medications regardless of the magnitude of BP elevation. Because only a minority of participants were prescribed mineralocorticoid receptor antagonists (e.g., spironolactone or eplerenone), it could be argued that most patients were not on “maximal” medical therapy. Because medication non-adherence can lead to misdiagnosis of RH, we assessed adherence both by self-report and objectively using MEMS technology. Medication adherence appeared excellent in both groups, both before and after the intervention, but urine levels of antihypertensive

medications and their metabolites were not measured, so adherence could not be confirmed with certainty. Although ambulatory BPs were not used to rule out ‘white coat’ hypertension (e.g., participants on 3 drugs and a clinic BP $\geq 130/80$ mm Hg but an average 24-hour BP $< 125/75$ mmHg), a post-hoc analysis revealed that only 2 patients with elevated clinic BP may have had apparent RH due to the white coat phenomenon. All TRIUMPH participants were considered to have essential hypertension by their referring physicians, but it is possible that some had unrecognized secondary hypertension, especially primary hyperaldosteronism. A recent analysis in patients with RH demonstrated a high prevalence of renin-independent aldosterone production, regardless of the aldosterone-renin ratio⁴⁹. Although both groups were similar on most baseline clinical and demographic characteristics, the SEPA group included more African Americans compared with C-LIFE; however, results were unchanged when race was included as a covariate in our analyses. The 4-month intervention might have been too short to produce clinically meaningful changes in some CVD biomarkers, and the extent to which improved health behaviors can be sustained without a structured program and ongoing accountability is uncertain. While the TRIUMPH lifestyle intervention is effective in reducing BP and improving important CVD biomarkers, there is an expense to delivering the intervention within the context of established cardiac rehabilitation programs, and the cost-effectiveness of the intensive lifestyle program will need to be evaluated.

CONCLUSIONS

Among a sample of 140 patients with RH, an intensive 4-month program of exercise and dietary modification as adjunctive therapy delivered in a cardiac rehabilitation setting resulted in improved BP control in patients with RH. These findings suggest that successful lifestyle changes can best be accomplished by a comprehensive, multidisciplinary team of health professionals within the established infrastructure of cardiac rehabilitation. Results of the TRIUMPH study suggest that policymakers should consider RH as a new indication for cardiac rehabilitation with appropriate coverage by governmental agencies and private insurers.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Abbreviations and Acronyms:

BP	Blood pressure
BRS	Baroreflex sensitivity
CI	Confidence interval
C-LIFE	Center-based lifestyle intervention
DBP	Diastolic blood pressure
DDD	Daily defined dose
HF-HRV	High frequency heart rate variability
PWV	Pulse wave velocity
RH	Resistant hypertension
SBP	Systolic blood pressure
SEPA	Standard education and physician advice

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CLINICAL PERSPECTIVE**WHAT IS NEW?**

- The effect of lifestyle modifications on blood pressure in patients with resistant hypertension is not known.
- This is the first study to show that a structured and supervised program of lifestyle modifications can reduce blood pressure and improve important CVD biomarkers in patients with resistant hypertension.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Patients with resistant hypertension can successfully adopt a healthy eating plan, lose weight, and improve their aerobic fitness by participating in a structured program delivered in a cardiac rehabilitation setting.
- Lifestyle modification including adoption of the DASH diet with caloric restriction and reduced salt consumption and regular aerobic exercise can significantly lower blood pressure and improve CVD biomarkers.

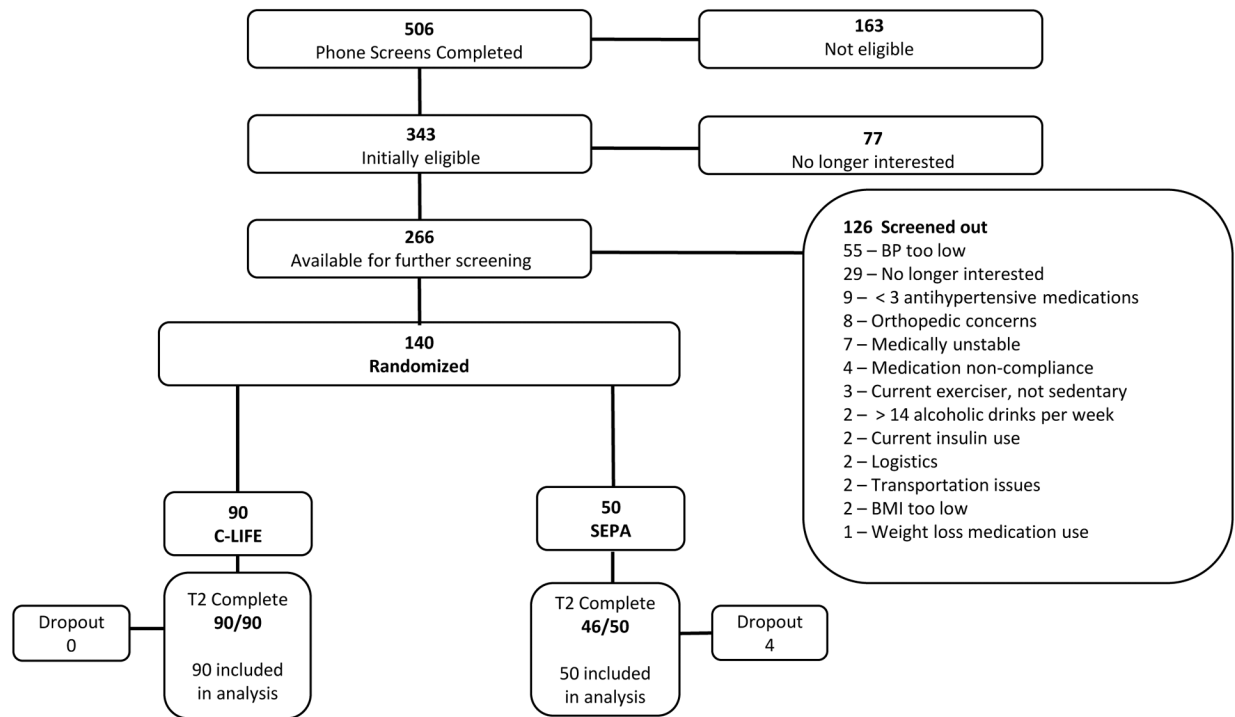


Figure 1:
CONSORT chart of trial enrollment.

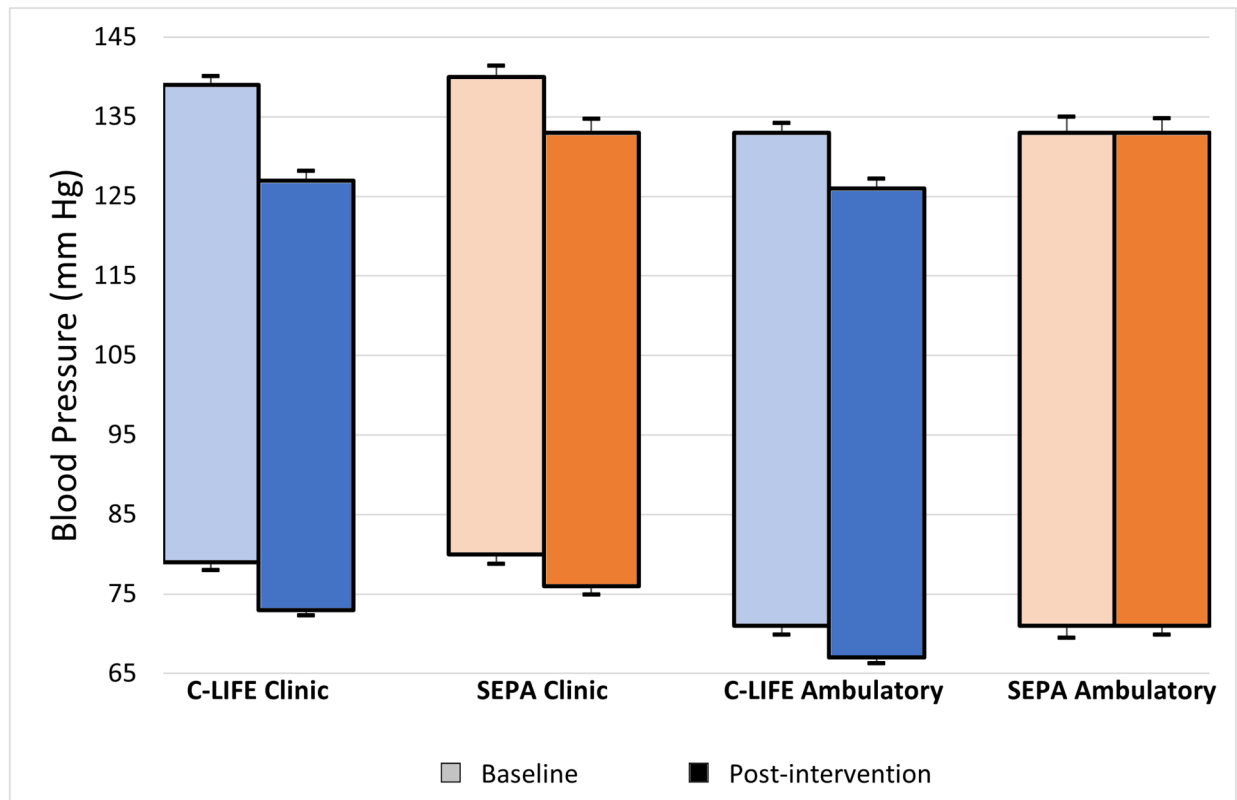


Figure 2: Mean Baseline (Pre-) and Post-Treatment Clinic and 24-hour Ambulatory Blood Pressure by Treatment Group

Baseline (pre-treatment) and post-treatment unadjusted systolic and diastolic clinic blood pressure (left) and systolic and diastolic ambulatory blood pressure (right) by treatment group (C-LIFE in blue; SEPA in orange). Error bars represent standard errors.

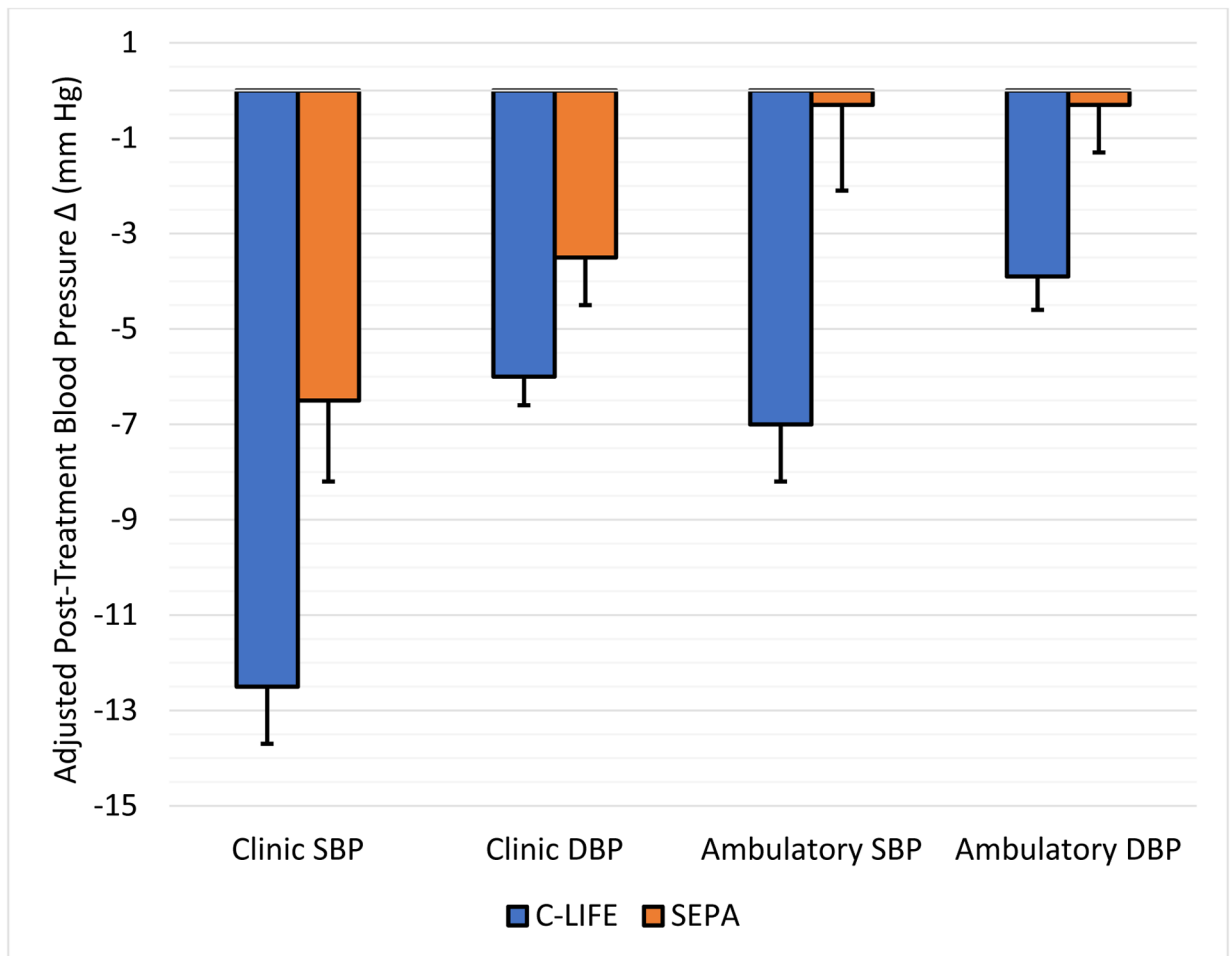


Figure 3: Mean Changes in Clinic and Ambulatory Blood Pressure from Baseline by Treatment Group

Adjusted mean systolic and diastolic blood pressure changes from baseline to post-treatment by treatment group. C-LIFE (blue) and SEPA (orange) plotted separately. Error bars represent standard errors.

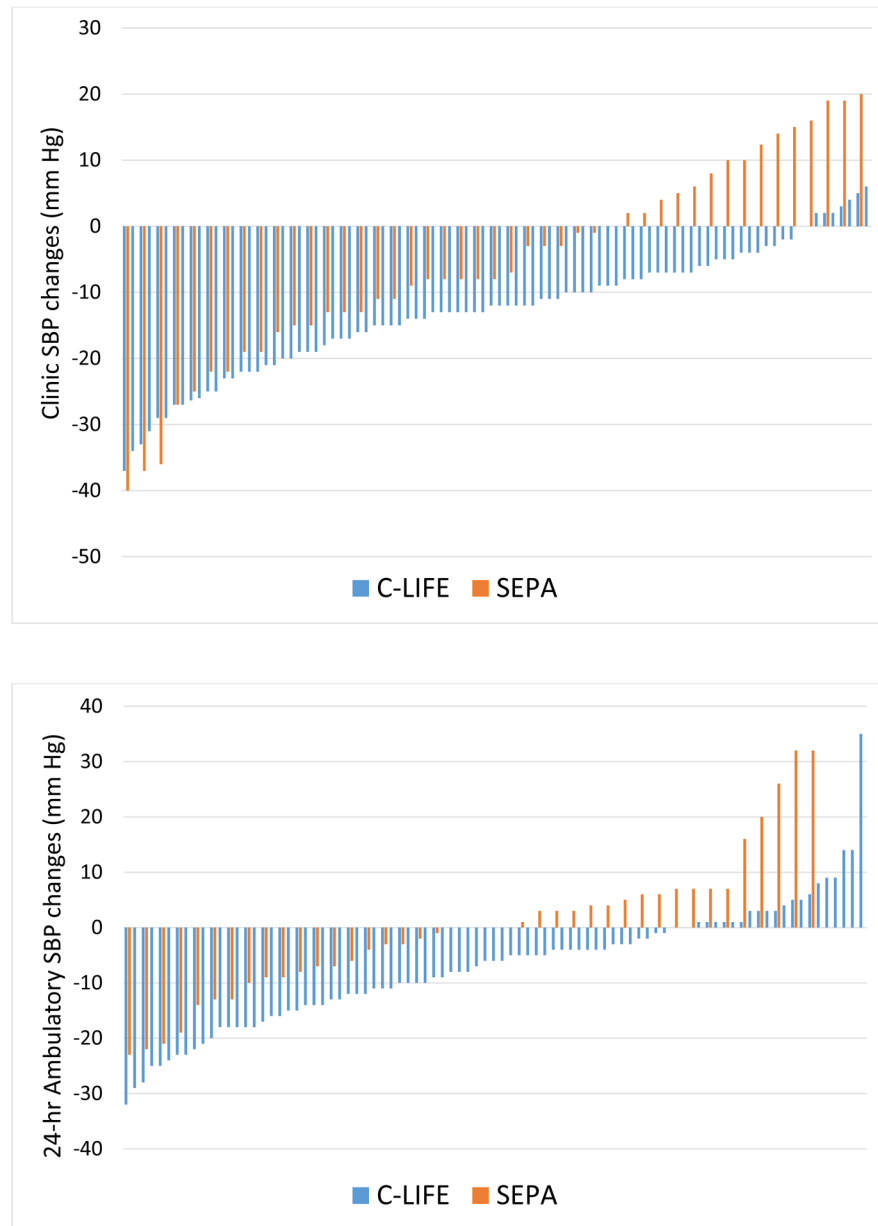


Figure 4: Individual changes in Clinic and 24-hour Ambulatory SBP for C-LIFE and SEPA groups

Individual systolic blood pressure (SBP) changes from baseline to post-treatment for clinic SBP (top) and ambulatory SBP (bottom) for C-LIFE (blue) and SEPA (orange).

Table 1.

Background and clinical characteristics of the TRIUMPH cohort.

Variable	C-LIFE (N = 90)	SEPA (N = 50)	Total Cohort (N = 140)
Age (yrs)	62 (9)	63 (9)	63 (9)
Sex (N, % Female)	43 (48%)	24 (48%)	67 (48%)
Race / Ethnicity			
Black	47 (52%)	35 (70%)	82 (59%)
White	39 (43%)	11 (22%)	50 (36%)
Asian	1 (1%)	1 (2%)	2 (1%)
Native American	0 (0%)	2 (4%)	2 (1%)
Hispanic	1 (1%)	1 (2%)	2 (1%)
Other	2 (2%)	0 (0%)	2 (1%)
Education Level			
High School Or Less	6 (7%)	6 (12%)	12 (8%)
Some College	30 (33%)	13 (26%)	43 (31%)
≥ College Graduate	54 (60%)	31 (62%)	85 (61%)
Annual Household Income (\$)			
Preferred Not To Respond	3 (3%)	1 (2%)	4 (3%)
<30k	7 (8%)	5 (10%)	12 (9%)
30–45k	13 (14%)	6 (12%)	19 (14%)
45–60k	13 (14%)	5 (10%)	18 (13%)
60–75K	11 (12%)	4 (8%)	15 (11%)
75–100k	15 (17%)	15 (30%)	30 (21%)
100–200k	21 (23%)	9 (18%)	30 (21%)
>200k	7 (8%)	5 (10%)	12 (9%)
Work Status			
Full-Time	29 (32%)	18 (36%)	47 (34%)
Part-Time	15 (17%)	8 (16%)	23 (16%)
Retired / Not Employed	46 (51%)	24 (48%)	70 (50%)
Marital Status			
Married/ Co-Habiting	58 (64%)	42 (84%)	100 (71%)
Single, Never Married	16 (18%)	1 (2%)	17 (12%)
Divorced/Separated	10 (11%)	7 (14%)	17 (12%)
Widowed	6 (7%)	0 (0%)	6 (4%)
Clinical Characteristics			
Clinic SBP (mm Hg)	139 (10)	140 (10)	139 (10)
Clinic DBP (mm Hg)	79 (9)	80 (8)	79 (9)
24-hour Ambulatory SBP (mm Hg)	133 (11)	133 (14)	133 (12)
24-hour Ambulatory DBP (mm Hg)	71 (10)	71 (11)	71 (10)

Variable	C-LIFE (N = 90)	SEPA (N = 50)	Total Cohort (N = 140)
Daytime Ambulatory SBP (mm Hg)	136 (12)	138 (15)	137 (13)
Daytime Ambulatory DBP (mm Hg)	74 (10)	74 (11)	74 (11)
Nighttime Ambulatory SBP (mm Hg)	123 (15)	123 (16)	123 (15)
Nighttime Ambulatory DBP (mm Hg)	64 (11)	62 (10)	63 (11)
Weight (lb)	230 (45)	231 (41)	230 (44)
Body Mass Index (kg/m ²)	36 (6)	36 (5)	36 (6)
Diabetes	33 (37%)	11 (22%)	44 (31%)
Current Smoker	1 (1%)	5 (10%)	6 (4%)
History of MI	6 (7%)	1 (2%)	6 (4%)
History of PCI	2 (2%)	1 (2%)	7 (5%)
History of CABG	6 (7%)	1 (2%)	3 (2%)
History of Stroke	4 (4%)	3 (6%)	7 (5%)
History of CKD	18 (20%)	16 (32%)	34 (24%)
eGFR (mL/min/1.73m ²)	78 (17)	76 (21)	77 (19)
Total Cholesterol (mg/dL)	174 (43)	179 (42)	176 (43)
LDL Cholesterol (mg/dL)	98 (38)	100 (34)	99 (37)
HDL Cholesterol (mg/dL)	50 (17)	55 (17)	52 (17)
VLDL Cholesterol (mg/dL)	26 (14)	24 (10)	25 (13)
Triglycerides (mg/dL)	130 (72)	120 (51)	127 (65)
Glucose (mg/dL)	111 (22)	110 (24)	110 (23)
Insulin (mIU/L)	17 (8)	17 (11)	17 (9)
Statins / Cholesterol Lowering Medications	47 (52%)	28 (56%)	75 (54%)
Peak VO ₂ (ml/kg/min)	17.9 (5)	16.6 (4)	17.4 (5)
ETT Duration (min)	7.4 (2)	6.8 (2)	7.2 (2)
DASH Eating Plan	3.7 (1)	3.4 (1)	3.5 (1)
ASCVD 10-year Risk Score (%)	13.4 (8)	13.2 (8)	13.4 (8)
Duration of Hypertension, years	22.2 (12.5)	22.3 (10.6)	22.2 (11.8)
Antihypertensive Medications			
Antihypertensive Medication Burden, Daily Defined Dose Sum	5.1 (2)	4.9 (2)	5.0 (2)
Number of Antihypertensive Medications	3.5 (0.7)	3.4 (0.8)	3.5 (0.7)
Thiazide or Thiazide-Like Diuretics	64 (71%)	36 (72%)	100 (71%)
Loop Diuretics	11 (12%)	6 (12%)	17 (12%)
Mineralocorticoid Receptor Antagonists	22 (24%)	7 (14%)	29 (21%)
Other Potassium Sparing Diuretics	9 (10%)	6 (12%)	15 (11%)
Beta-Blockers	66 (73%)	35 (70%)	101 (72%)
Angiotensin Receptor Blockers	47 (52%)	35 (70%)	73 (52%)
Calcium Antagonists	71 (79%)	37 (74%)	108 (77%)
ACE-Inhibitors	33 (37%)	23 (46%)	56 (40%)
Alpha Adrenergic Receptor Blockers	0 (0%)	2 (4%)	2 (1%)

Variable	C-LIFE (N = 90)	SEPA (N = 50)	Total Cohort (N = 140)
Central Alpha Adrenergic Receptor Agonists	3 (3%)	4 (8%)	7 (5%)
Direct Vasodilators	9 (10%)	2 (4%)	11 (8%)

Note: ASCVD = atherosclerotic cardiovascular disease; CABG = coronary artery bypass grafting; DASH = Dietary Approaches to Stop Hypertension; MI = myocardial infarction; PCI = percutaneous coronary intervention; ETT = Exercise treadmill test; VO₂peak = peak oxygen consumption

Table 2:

Mean Changes (with 95% Confidence Intervals) from Baseline (Pre-) to Post-intervention (with 95% Confidence intervals) for Physical Activity, Aerobic Fitness, Dietary Variables and Urinary Biomarkers in C-LIFE vs. SEPA.

Variable	C-LIFE			SEPA			Group Contrast, P-value
	Pre (SD)	Adjusted Post	Adjusted Δ	Pre (SD)	Adjusted Post	Adjusted Δ	
Physical Activity / Aerobic Fitness/ Weight							
Actigraphy, Total Steps	10,157 (2450)	11,216 (10,784, 11,648)	971 (568, 1374)	10,497 (2701)	10,104 (9452, 10,756)	14 (-580, 608)	.024
ETT Peak VO ₂ (ml/kg/min)	17.9 (4.5)	20.0 (19.3, 20.6)	2.40 (1.75, 3.05)	16.6 (4.4)	17.9 (16.8, 18.9)	0.57 (-0.41, 1.54)	.002
ETT Peak VO ₂ (L/min)	1.9 (0.6)	1.9 (1.9, 2.0)	0.10 (0.05, 0.16)	1.7 (0.6)	1.8 (1.7, 1.9)	-0.01 (-0.10, 0.07)	.021
ETT Duration (min)	7.4 (2.1)	9.3 (9.0, 9.6)	2.09 (1.8, 2.4)	6.8 (2.4)	7.6 (7.1, 8.1)	0.58 (0.11, 1.05)	<.001
Weight (lbs)	230 (45)	215 (213, 217)	-15.3 (-17.2, -13.3)	231 (41)	221 (219, 224)	-8.5 (-11.4, -5.6)	<.001
Body Mass Index (kg/m ²)	35.9 (6.0)	33.6 (33.3, 34.0)	-2.34 (-2.65, -2.02)	36.1 (5.2)	34.7 (34.2, 35.1)	-1.26 (-0.80, -1.72)	<.001
Dietary Data							
Calories (kcal/day)	1811 (480)	1666 (1572, 1761)	-187 (-282, -92)	1912 (589)	1616 (751, 1482)	-226 (-377, -75)	.281
Protein (g/day)	80.6 (24)	78.7 (74.2, 83.2)	-3.0 (-1.4, 7.5)	83.4 (25)	77.1 (70.7, 83.5)	-5.0 (-2.0, 12.0)	.097
Carbohydrates (g/day)	205 (61)	195 (183, 207)	-16.2 (-28.3, 4.1)	219 (80.2)	179 (162, 197)	-29.0 (-47.9, -10.0)	.021
Fiber (g/day)	19.2 (7.1)	20.9 (19.3, 22.4)	2.3 (0.7, 3.8)	17.3 (5.3)	18.1 (15.8, 20.4)	0.2 (-2.7, 2.4)	.040
Magnesium (mg/day)	306 (105)	316 (297, 334)	6.2 (-10.3, 22.7)	295 (80)	283 (256, 310)	-35.3 (-59.8, -10.9)	.008
Potassium (mg/day)	2798 (790)	2846 (2702, 2990)	66 (-68, 200)	2741 (796)	2556 (2350, 2761)	-243 (-411, -45)	.014
Sodium (mg/day)	3173 (879)	2975 (2805, 3145)	-221 (-393, -49)	3205 (893)	2874 (2631, 3117)	-277 (-547, -8)	.244
Calcium (mg/day)	815 (316)	771 (714, 828)	-41.4 (-99.2, 16.4)	803 (311)	751 (670, 833)	-61.1 (-158.3, 36.4)	.418
DASH Eating Plan Score	3.7 (1.2)	4.0 (3.8, 4.3)	0.6 (0.3, 0.8)	3.4 (1.1)	3.8 (3.5, 4.2)	0.3 (-0.1, 0.6)	.262
Vegetable Servings/day	3.7 (1.8)	4.4 (4.0, 4.8)	0.6 (0.2, 1.1)	4.0 (2.0)	3.3 (2.7, 3.9)	-0.5 (-1.1, 0.0)	.004
Fruit Servings/day	3.3 (2.2)	4.0 (3.5, 4.5)	0.7 (0.2, 1.2)	3.0 (1.9)	3.4 (2.7, 4.1)	0.2 (-0.5, 1.0)	.350
Whole Grain Servings/day	1.0 (0.9)	1.2 (1.0, 1.4)	0.2 (0.0, 0.4)	0.8 (0.8)	1.0 (0.7, 1.3)	0.1 (-0.3, 0.4)	.251
Nuts and Legumes/day	1.8 (2.1)	1.9 (1.3, 2.4)	0.2 (-0.4, 0.7)	1.4 (1.5)	2.1 (1.3, 2.9)	0.4 (-1.3, 0.4)	.606
Sweets Servings/week	18.5 (12.2)	15.5 (13.2, 17.8)	-5.6 (-7.9, -3.3)	25.6 (20.7)	15.5 (12.1, 19.0)	-6.3 (-9.6, -3.0)	.987
Total Fat (%)	36.1 (6.2)	34.7 (33.2, 36.1)	-1.6 (-3.0, -0.2)	36.7 (4.4)	35.6 (33.6, 37.7)	-0.8 (-2.8, 1.1)	.407
Saturated Fat (%)	11.3 (2.9)	10.3 (9.7, 10.9)	-1.0 (-1.6, -0.4)	11.2 (1.9)	10.9 (10.0, 11.7)	-0.4 (-1.2, 0.4)	.167

Variable	G-LIFE			SEPA			Group Contrast, P-value
	Pre (SD)	Adjusted Post	Adjusted Δ	Pre (SD)	Adjusted Post	Adjusted Δ	
Phosphorus (mg/day)	1288 (371)	1242 (1176, 1308)	-52.6 (-117.6, 12.4)	1299 (354)	1209 (1113, 1303)	-90.8 (-186.9, 5.3)	.359
Urinary Electrolytes							
Urine Sodium Excretion, mg/day	2741 (1007)	2473 (132)	-308 (-567, -50)	2857 (1423)	3023 (240)	187 (-288, 662)	.007
Urine Potassium Excretion, mg/day	1810 (606)	2178 (2048, 2308)	416 (286, 546)	1678 (742)	2064 (1847, 2281)	292 (96, 488)	.382

Note: ETT = Exercise Treadmill Test; DASH = Dietary Approaches to Stop Hypertension Diet. Dietary data were derived from the Dietary recall and Food Frequency Questionnaire.

Table 3.

Mean Baseline to Post-treatment Changes in CVD and Metabolic Biomarker Values (with 95% Confidence Intervals) Following C-LIFE and SEPA Interventions

Variable	C-LIFE			SEPA		Group Contrast, P-value	
	Pre (SD)	Adjusted Post	Adjusted Δ	Pre (SD)	Adjusted Post		Adjusted Δ
CVD Biomarkers							
BRS (msec/mmHg)	6.0 (3.4)	7.4 (6.4, 8.3)	2.3 (1.3, 3.3)	5.5 (3.2)	5.3 (3.9, 6.7)	-1.1 (-2.5, 0.3)	.003
HF-HRV (ms ²)	5.7 (1.1)	6.1 (5.9, 6.3)	0.4 (0.2, 0.6)	5.7 (1.3)	5.7 (5.4, 6.0)	-0.2 (-0.5, 0.1)	.025
Flow Mediated Dilatation, %	2.1 (2.5)	2.7 (2.1, 3.4)	0.3 (-0.3, 1.0)	2.0 (2.1)	1.2 (0.1, 2.3)	-1.4 (-2.5, -0.3)	.022
LVM (g/m ^{2.7})	47.5 (12.6)	46.7 (44.6, 46.8)	-2.9 (-3.7, -2.0)	49.5 (13.4)	46.5 (45.1, 47.9)	-2.3 (-3.7, -0.8)	.596
Relative Wall Thickness (g/m ²)	0.5 (0.1)	0.44 (0.43, 0.45)	-0.03 (-0.04, -0.01)	0.5 (0.1)	0.45 (0.43, 0.46)	-0.01 (-0.03, 0.01)	.129
PWV (m/sec)	9.8 (4.2)	9.7 (9.2, 10.2)	0.16 (-0.40, 0.72)	10.1 (2.3)	10.1 (9.3, 10.9)	0.14 (-0.72, 1.0)	.958
Metabolic Biomarkers							
hsCRP (mg/L)	3.7 (3.0)	3.0 (2.6, 3.5)	-0.72 (-1.14, -0.30)	3.8 (2.6)	3.5 (2.8, 4.1)	-0.23 (-0.90, 0.44)	.550
Total Cholesterol (mg/dL)	174 (43)	168 (163, 173)	-8.4 (-14.3, -2.6)	179 (42)	168 (160, 176)	-9.6 (-18.6, -0.7)	.855
HDL-cholesterol (mg/dL)	50.3 (17)	53 (51, 54)	0.5 (-1.3, 2.3)	55 (17)	53 (51, 56)	1.4 (-1.5, 4.2)	.574
LDL-cholesterol (mg/dL)	97.6 (38)	94 (90, 98)	-4.8 (-9.5, -0.5)	100 (34)	92 (86, 99)	-7.8 (-15.1, -0.6)	.872
VLDL-cholesterol (mg/dL)	26 (14)	22 (20, 23)	-3.9 (-5.6, -2.2)	24 (10)	22 (20, 24)	-3.5 (-6.1, -1.0)	.870
Triglycerides (mg/dL)	130 (72)	109 (100, 117)	-19.5 (-27.9, -11.0)	120 (51)	110 (98, 122)	-17.7 (-30.5, -4.9)	.880
Glucose (mg/dL)	111 (22)	104 (100, 107)	-6.8 (-10.3, -3.4)	112 (35)	105 (100, 110)	-7.4 (-12.6, -2.3)	.529
Insulin (uIU/ml)	17 (8)	12 (11, 13)	-5.1 (6.5, -3.7)	17 (11)	13 (11, 15)	-3.6 (-5.8, -1.5)	.469
HbA1c (%)	6.2 (0.7)	5.9 (5.8, 6.0)	-0.28 (-0.40, -0.17)	6.2 (1.0)	6.0 (5.8, 6.1)	-0.30 (-0.46, -0.14)	.212

Note: Overall intervention group contrast of CVD Biomarker global score for C-LIFE and SEPA: $P = .027$.

BRS = Baroreflex sensitivity; HF-HRV = high frequency heart rate variability; LVM = left ventricular mass index; PWV = pulse wave velocity; HbA1c = hemoglobin A1c